SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME

COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5 Total

Marks:50 Annexure A

Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools ***(2)***	Analysis of Results ***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. ***Signature of the student:***

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understandi ng; limited or partial integration	Poor understanding; unable to integrate concepts
Experime ntal Design	30	Well-planned, systematic implementation with optimal solution	Correct implementatio n with minor issues	Basic implementati on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documenta tio n	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentati on with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child A B C 5-6 years: 18 22 20

7-8 years: 2 28 40 9-10 years: 20 10 40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences. Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored
P1 Player A 12
P2 Player B 15
P3 Player C 14
P4 Player D 10
P5 Player E 18
P6 Player F 20
P7 Player G 22
P8 Player H 13
P9 Player I 11
P10 Player J 35
P11 Player K 16
P12 Player L 17
P13 Player M 19
Player ID Player Name Points Scored
P14 Player N 21
P15 Player O 23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**. (10 Marks)

Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes

middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*. (b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Answers – Assessment Tool 1 (CO3, CSA16)

Q1. Chi-Square Test of Independence & Covariance/Correlation

Data (observed frequencies):

Age of child	Photo A	Photo B	Photo C	Row Total
5–6 years	18	22	20	60
7–8 years	2	28	40	70
9–10 years	20	10	40	70
Column Total	40	60	100	200

Hypotheses: H0: Age and photograph preference are independent. H1: Age and photograph preference are associated.

Expected frequency E_{ij} = (row total × column total) / grand total. This gives the expected table:

Age of child	Photo A	Photo B	Photo C
5–6 years	12.0	18.0	30.0
7–8 years	14.0	21.0	35.0
9–10 years	14.0	21.0	35.0

$\chi^2 = \Sigma (O-E)^2/E$ = **29.60**, with df = (3−1)(3−1) = 4. At $\alpha = 0.05$, critical $\chi^2_{0.05,4} = 9.49$ ($p \approx 5.9\times10^{-6}$). Since 29.60 > 9.49, we **reject H0**: there is a statistically significant relationship between a child's age and their photograph preference.

What is wrong with this study:

• The three age groups have unequal and fairly small sample sizes (60, 70, 70), which limits the precision and generalizability of the conclusions.
• Children were not (evidently) randomly sampled or randomly assigned to age groups — the design is observational, so the association found cannot be interpreted as a causal effect of age. • Each child appears to have made only a single choice with no replication or control condition, so response consistency cannot be checked.
• The order in which the three photographs were presented is not controlled, which could introduce order/position bias.
• Other confounding variables (gender, cultural background, familiarity with the adults shown) are not accounted for.

Covariance and correlation between B and C (using the three age-group counts as paired observations):

Sample covariance: cov(B, C) = −**20.00**
Sample correlation: cor(B, C) = −**0.189** (a weak negative linear relationship)

Full sample covariance matrix (rows/columns in order A, B, C):

	A	**B**	**C**
A	97.33	-74.00	-46.67
B	-74.00	84.00	-20.00
C	-46.67	-20.00	133.33

Full sample correlation matrix:

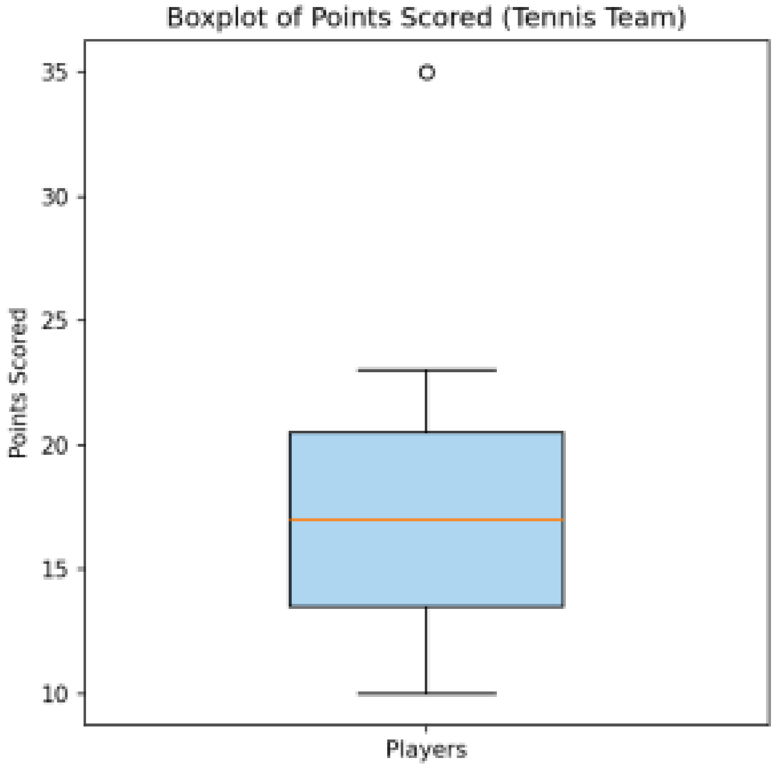
	A	**B**	**C**
A	1.000	-0.818	-0.410
B	-0.818	1.000	-0.189
C	-0.410	-0.189	1.000

Q2. Boxplot for Outlier Detection – Tennis Points Scored

Approach: To visualize the distribution and flag outliers, the coach can construct a boxplot of the 15 players' points using the five-number summary (minimum, Q1, median, Q3, maximum) and the 1.5×IQR rule:

1. Sort the 15 scores: 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35. 2.
Compute Q1 (25th percentile) = 13.5, Median = 17.0, Q3 (75th percentile) = 20.5. 3.
IQR = Q3 − Q1 = 7.0.
4. Lower fence = Q1 − 1.5×IQR = 3.0; Upper fence = Q3 + 1.5×IQR = 31.0.
5. Any point outside [3.0, 31.0] is plotted individually as an outlier; the whiskers extend only to the most extreme data points within the fences.

Mean = 17.73, Median = 17.0.



Result: Player J (P10, 35 points) lies above the upper fence of 31.0 and is flagged as an **outlier** — a scoring performance well above the rest of the team. All other players fall within the whiskers (10–23), representing the team's typical scoring range.

Q3. Decision Tree Classifier – Loan Approval

The dataset has 15 records, 4 candidate attributes (Age, Income, Employment Status, Credit Score) and target class Loan Approval (Yes/No): 9 Yes, 6 No.

Root entropy: H(D) = −(9/15)log■(9/15) − (6/15)log■(6/15) = **0.971**

Information gain for each attribute (root split):

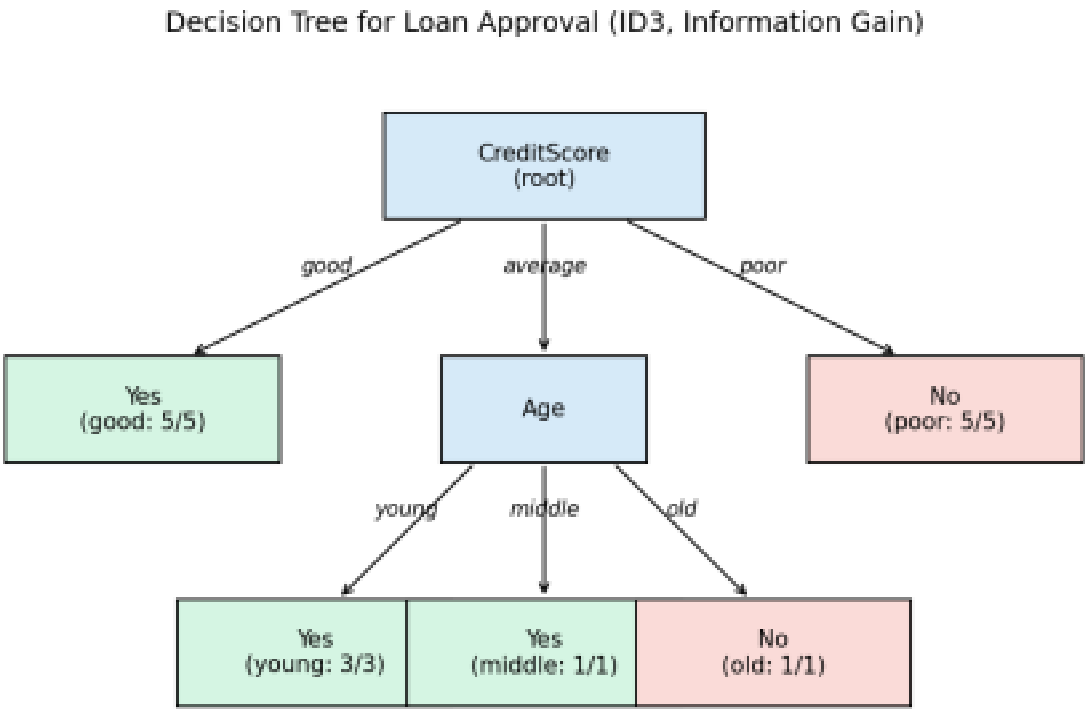
Attribute	Info(attr)(D)	Gain(attr)
Age	0.888	0.083
Income	0.260	0.711
Employment Status	0.501	0.470
Credit Score	0.241	0.730

Credit Score gives the highest information gain (0.730), so it is selected as the root node. Its branches: good (5 records, all Yes → pure leaf), poor (5 records, all No → pure leaf), and average (5 records: 4 Yes, 1 No, entropy = 0.722) which needs further splitting.

Splitting the Credit Score = average branch (5 records):

Attribute	Weighted Info	Gain
Age	0.000	0.722
Income	0.000	0.722
Employment Status	0.551	0.171

Age and Income both give a perfect (pure) split here; **Age** is chosen: young → 3/3 Yes, middle → 1/1 Yes, old → 1/1 No.



Resulting rules:

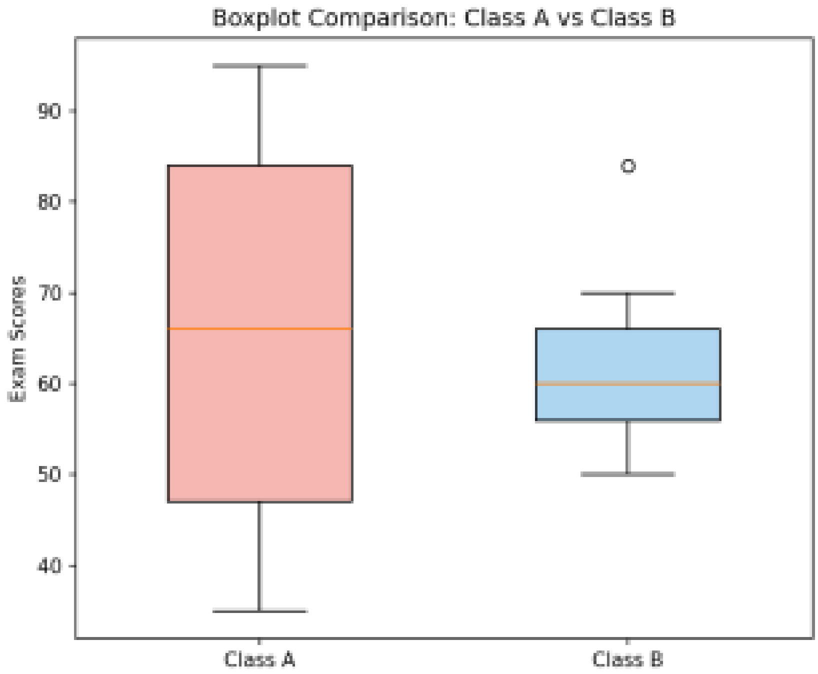
- IF Credit Score = good THEN Loan Approval = Yes
- IF Credit Score = poor THEN Loan Approval = No
- IF Credit Score = average AND Age = young THEN Loan Approval = Yes
- IF Credit Score = average AND Age = middle THEN Loan Approval = Yes
- IF Credit Score = average AND Age = old THEN Loan Approval = No

Q4. Comparing Class A and Class B Exam Scores

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84 (n = 9)
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50 (n = 9)
(Note: the pasted values in the question repeat each list twice with no separator; the distinct 9-value data set for each class is used above.)

Statistic	**Class A**	**Class B**
Mean	65.78	62.11
Median	66.0	60.0
Min – Max	35 – 95	50 – 84
Range	60	34

(i) Class A has the higher mean (65.78 vs 62.11), higher median (66.0 vs 60.0), and a much larger range (60 vs 34).



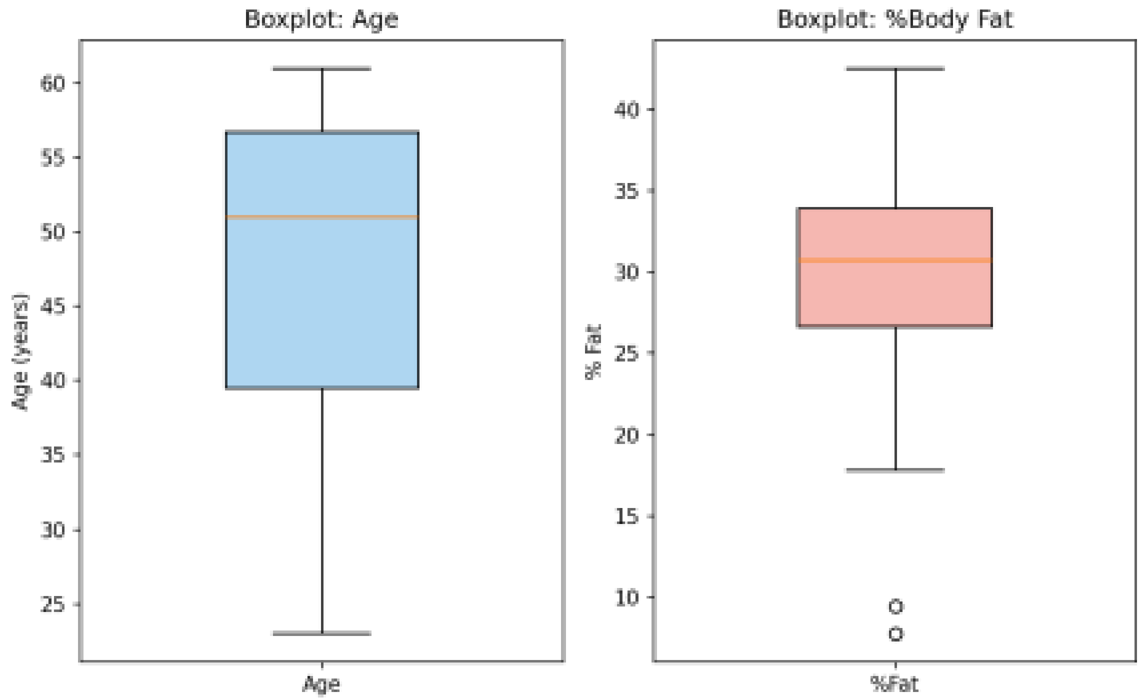
(ii) Inferences from the boxplot: Class A's box and whiskers are noticeably wider, showing far greater variability — some students scored very low (35) while others scored very high (95), suggesting a mixed-ability class or inconsistent performance. Class B's scores are far more tightly clustered around the low-to-mid 60s, indicating a more consistent/uniform class performance, though the value 84 falls outside Class B's upper fence (Q3 + 1.5×IQR = 81.0) and shows up as an **outlier**. Overall, Class A performed slightly better on average but far less consistently than Class B.

Q5. Age and %Body Fat – Descriptive Statistics & Visualizations

(a) Descriptive statistics (n = 18):

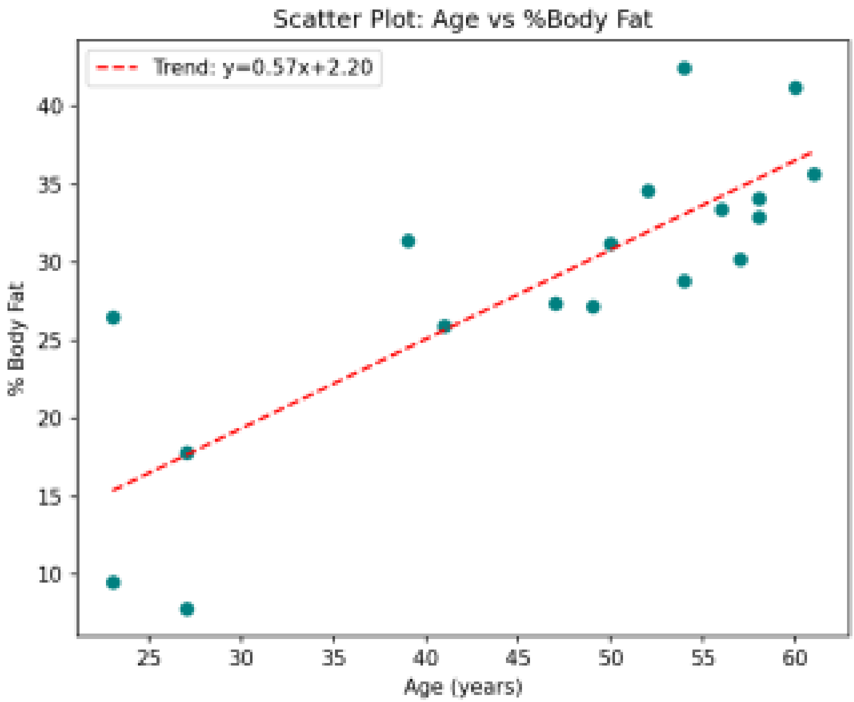
Variable	Mean	Median	Std. Deviation
Age (years)	46.44	51.0	13.22
% Fat	28.78	30.7	9.25

(b) Boxplots for Age and %Fat:

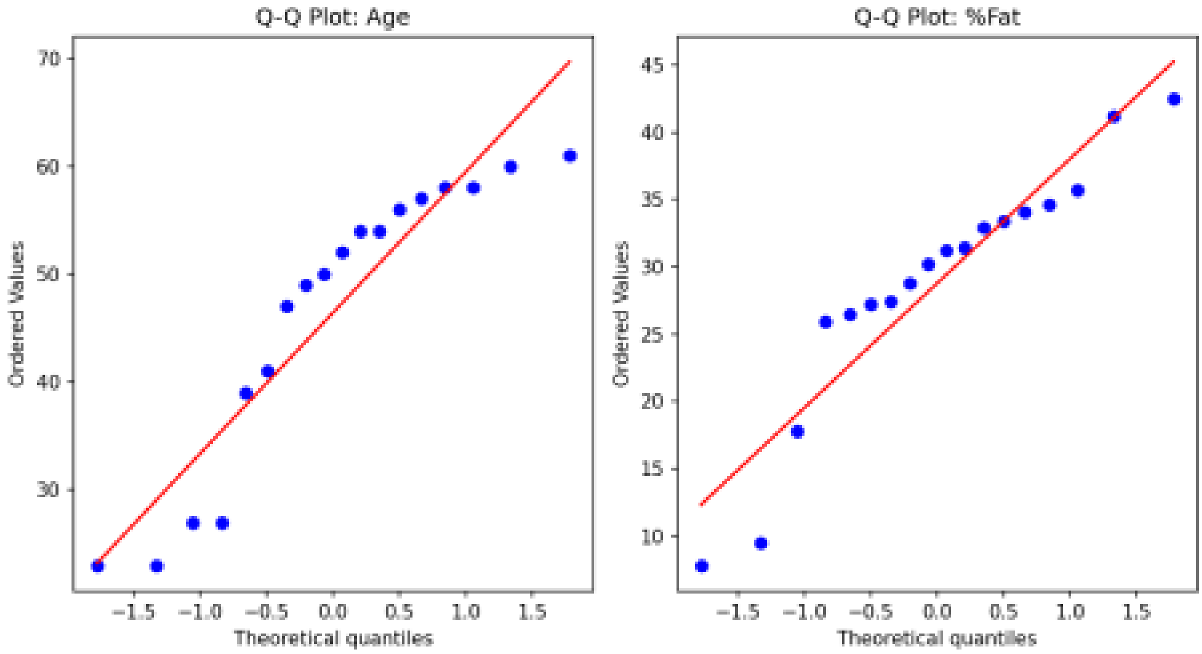


Both variables show a fairly symmetric spread with no extreme outliers; %Fat has a somewhat wider interquartile spread relative to its scale, reflecting more individual variability in body fat than in age.

(c) Scatter plot and Q–Q plots:



The scatter plot shows a clear positive linear trend (r = 0.818): %body fat tends to increase with age in this sample.



The Q–Q plots show both Age and %Fat points lying reasonably close to the reference line, suggesting neither variable departs strongly from normality — both are approximately normally distributed, supporting the use of parametric methods (e.g. Pearson correlation) computed above.